

J. Alex Baecher

Curriculum Vitae

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EDUCATION

Doctor of Philosophy, Interdisciplinary Ecology <i>University of Florida</i>	Aug 2024
Master of Science, Biology <i>Eastern Kentucky University</i>	Dec 2017
Bachelor of Science, Biology <i>University of Arkansas</i>	May 2014

APPOINTMENTS

Assistant Professor of Applied Ecology <i>Emory University</i> Department of Environmental Science	Aug 2026 – Present
Postdoctoral Fellow in Quantitative Ecology <i>Utah State University</i> Department of Rangeland Resources	July 2025 – July 2026
Postdoctoral Research Associate <i>University of Florida</i> Institute for Food and Agricultural Science	Aug 2024 – July 2025
Doctoral Preeminence Fellow <i>University of Florida</i> Department of Wildlife Ecology	Aug 2019 – Aug 2024
Conservation and Research Technician <i>Memphis Zoo</i> Department of Conservation and Research	Jan 2018 – July 2019
NSF REU Research Coordinator <i>Eastern Kentucky University</i> Department of Biology	May 2017 – Aug 2017
Graduate Teaching Assistant <i>Eastern Kentucky University</i> Department of Biology	May 2015 – Aug 2017
Research Assistant <i>University of Arkansas</i> Department of Biology	Mar 2012 – Mar 2015

FUNDING

Amount	Organization	Fund Name	Year
\$680,000	Emory University	Lab Start-up Fund	2026
\$130,000	US Geological Survey	Climate Adaptation Postdoctoral Fellows	2025
\$195,000	University of Florida	Doctoral Preeminence Fellowship Award	2019
\$4,000	University of Florida	Robin Nadeau Graduate Research Award	2019
\$5,000	National Science Foundation	Research Experience for Undergraduates	2013
\$5,000	National Science Foundation	Research Experience for Undergraduates	2012
\$1,100	Eastern Kentucky University	Marcia Athey Research Grant	2016
\$800	Eastern Kentucky University	Dept. Biology Travel Grant	2017
\$700	Society Freshwater Science	Instars Fellowship	2014
\$700	Society Freshwater Science	Instars Fellowship	2013
\$500	Soc. Study Amphibians & Reptiles	Grant-in-Aid of Herpetological Research	2016
\$500	Eastern Kentucky University	Dep. Biological Sciences Travel Stipend	2016
\$500	Eastern Kentucky University	Div. Natural Areas Research Grant	2016
\$500	University of Arkansas	Dept. Biology Travel Grant	2013

Total Funding: \$1,024,000

PUBLICATIONS

§Mentee co-author • Citations: **438** • h-index: **8**

Published

17. §Akshay, VA, **JA Baecher**, N Burkett-Cadena, J Thorson, §Y Sánchez, §Y Tavares, §A Bauer, RP Guralnick, LP Campbell. **2026**. *Linking hosts, landscapes, and climate to advance zoonotic arbovirus forecasting*. *Scientific Reports* [[html](#)] [[code](#)] [[preprint](#)]
16. Oliveira, BF, et al. [+21 co-authors including **JA Baecher**]. **2026**. *Species range shifts often speed ahead of their climatic niches*. *Proceedings of the National Academy of Sciences* [[html](#)] [[preprint](#)] [[code](#)] Press: [[Nature EcoEvo](#)] [[Conservation Science Partners](#)]
15. **Baecher, JA**, §VA Akshay, RP Guralnick, §AM Bauer, §Y Tavares, §Y Sánchez, LP Campbell. **2025**. *Toward ecological forecasting of West Nile virus in Florida: Insights from two decades of sentinel chicken monitoring*. *Science of the Total Environment* 1000:180308 [[html](#)] [[pdf](#)] [[code](#)] Press: [[Florida Museum](#)] [[The Microbiologist](#)]
14. Evans, LJ, **JA Baecher**, BR Scheffers. **2024**. *Invasion risk posed by the pet trade*. *Frontiers in Ecology and the Environment* [[html](#)] [[pdf](#)] [[data](#)]
13. **Baecher, JA**. **2024**. *The environmental filtering paradigm and non-filtering community assembly processes*. *Journal of Biogeography* 51(11):2109–2116 [[html](#)] [[pdf](#)]
12. Klinges, DH, **JA Baecher**, JJ Lembrechts, IMD Maclean, J Lenoir, C Greiser, M Ashcroft, LJ Evans, MR Kearney, J Aalto, LC Barrio, P De Frenne, J Guillemot, K Hylander, T Jucker, M Kopecky, M Luoto, M Macek, I Nijs, J Urban, L van den Brink, P Vangansbeke, J Von Oppen, J Wild, J Boike,

- R Canessa, M Noretto, A Rubtsov, J Sallo-Bravo, BR Scheffers. **2024**. *Proximal microclimate: moving beyond spatiotemporal scale to improve predicted biological responses*. *Global Ecology and Biogeography* 33(7):e1–16 [[html](#)] [[pdf](#)] [[code](#)] Award: [[Wiley's Top 10 Most Cited](#)]
11. Comte, L, R Bertrand, SE Diamond, LT Lancaster, ML Pinsky, BR Scheffers, **JA Baecher**, RMWJ Bandara, I-C Chen, JA Lawlor, NA Moore, BR Oliveira, J Murienne, J Rolland, MA Rubenstein, J Sunday, LM Thompson, F Villalobos, SR Weiskopf, J Lenoir. **2024**. *Bringing traits back in the equation: A roadmap to understand species redistribution*. *Global Change Biology* 30:e17271 [[html](#)] [[pdf](#)]
 10. Lawlor, JA, L Comte, G Grenouillet, J Lenoir, **JA Baecher**, RMWJ Bandara, R Bertrand, I-C Chen, SE Diamond, LT Lancaster, NA Moore, J Murienne, BF Oliveria, GT Pecl, ML Pinsky, J Rolland, MA Rubenstein, BR Scheffers, LM Thompson, B can Amerom, F Villalobos, SR Weiskopf, J Sunday. **2024**. *Mechanisms, detections, and impacts of species redistributions under climate change*. *Nature Reviews Earth & Environment* 5:351–368 [[html](#)] [[pdf](#)] *Impact Factor: 71.5 Altmetric: 134* (top 5%) [[Press: list of articles](#)]
 9. **Baecher, JA**, SA Johnson, EA Roznik, BR Scheffers. **2023**. *Experimental evaluation of how invasions and climate change interact to alter the vertical assembly of an amphibian community*. *Journal of Animal Ecology* 92(4):875–888 [[html](#)] [[pdf](#)] [[code](#)] Press: [[Nature Climate Change](#)] [[Florida Today](#)] [[Nature Climate Change](#)]
 8. Basham, EW, **JA Baecher**, DH Klings, BR Scheffers. **2022**. *Vertical stratification patterns of tropical forest vertebrates: A meta-analysis*. *Biological Reviews* 98(1):99–114 [[html](#)] [[pdf](#)] [[code](#)]
 7. Basham, EW, M González-Pinzón, A Romero-Marcucci, N Carl, **JA Baecher**, BR Scheffers. **2022**. *Large, old trees define the vertical, horizontal, and seasonal distributions of a poison frog*. *Oecologia* 199:257–269 [[html](#)] [[pdf](#)] [[code](#)]
 6. Swartwout, M, PN Vogrinc, **JA Baecher**, CS Kross, JD Willson. **2020**. *Prey Size and Growth Rate do not Influence Trophic Morphology of Juvenile Water Snakes Nerodia sipedon*. *Herpetologica* 76(1):53–60 [[html](#)] [[pdf](#)]
 5. **Baecher, JA**, SE Jones, SC Richter. **2020**. *Tail bifurcation in two species of Desmognathus salamander (Caudata: Plethodontidae) in southeastern Kentucky, USA*. *Herpetological Bulletin* 150:29–30 [[html](#)] [[pdf](#)]
 4. Hinkson, KM, **JA Baecher**, S Poo. **2019**. *Cryopreservation and hormonal induction of spermic urine in the smooth-sided toad Rhaebo guttatus*. *Cryobiology* 89:109–111 [[html](#)] [[pdf](#)] Press: [[The Commercial Appeal](#)] [[USA Today](#)]
 3. **Baecher, JA**, SC Richter. **2018**. *Environmental gradients in an old-growth Appalachian Forest predict fine-scale distribution, co-occurrence, and abundance of woodland salamanders*. *Ecology and Evolution* 8(24):12940–12952 [[html](#)] [[pdf](#)] [[code](#)]
 2. **Baecher, JA**, PH Vogrinc, JC Guzy, CS Kross, JD Willson. **2017**. *Herpetofaunal Communities in Restored and Unrestored Remnant Tallgrass Prairie and Associated Wetlands in Northwest Arkansas*. *Wetlands* 38:157–168 [[html](#)] [[pdf](#)]
 1. **Baecher, JA**, PH Vogrinc, JC Guzy, JC Neal, JD Willson. **2014**. *Lithobates areolatus, Predation*. *Herpetological Review* 45(4):681–682 [[pdf](#)]

Forthcoming

6. **Baecher, JA**, D Klinges, LJ Evans, CA Romagosa, R Fletcher, BR Scheffers. *In revision*. *Isolating what limits the spread of an invasive predator*. **Journal of Applied Ecology** [[code](#)] [[preprint](#)]
5. Thompson, LM, et al. [+21 co-authors including **JA Baecher**]. *In review*. *The RAD (Resist-Accept-Direct) management of species redistribution under climate change*. **Frontiers in Ecology and the Environment**
4. Klinges, DH, **JA Baecher**, L Soifer, BR Scheffers. *In review*. *Land modification defines climate change as experienced by life on earth*. **Science**
3. Pittman, C, et al., [+12 co-authors including **JA Baecher**]. *Submitted*. *Connecting living and preserved collections: Standardizing and integrating biological data to cultivate information sharing*. **Biodiversity Informatics**
2. **Baecher, JA**, D Klinges, LJ Evans, BR Scheffers. *In prep*. *Informing species range-shifts with global climate and habitat connectivity*. Prepared for **One Earth**
1. **Baecher, JA**, E Stuber, TJ Clark-Wolf. *In prep*. *Mapping range shifts with a novel integrated spatiotemporal N-mixture Bayesian model*. **Methods in Ecology and Evolution**

PROFESSIONAL ACTIVITIES

Peer Review

Total Requested: 38 | **Total Reviewed:** 25

Biological Conservation, Conservation Science and Practice, Ecography, Ecology, Ecology & Evolution, Frontiers in Ecology and Environment, Herpetological Bulletin, Herpetological Conservation Biology, Herpetological Review, Hydrobiologia, Natural Areas Journal, Nature Ecology & Evolution, Oecologia, Scientific Reports, Wildlife Society Bulletin

Professional Membership

Ecological Society of America	2019 – Present
International Association of Landscape Ecologists – North America	2020 – Present
Climate Adaptation Science Center (SE & SW)	2019 – Present
Southeast RISCC	2022 – Present
American Society of Ichthyologists and Herpetologists	2014 – 2021
Southeastern Partners in Amphibian and Reptile Conservation	2014 – 2021
Society of Freshwater Ecology	2012 – 2014

Working Groups

BioShifts (Member, Group Leader)	2022 – Present
DirtClim (Core team)	2019 – Present
National Synthesis for Ecological Analysis and Synthesis (Member)	2025 – Present
ResearchR (President of Scientific Integrity)	2025 – Present
ZooMu Network (Member, Group Leader)	2024 – Present
USGS Climate Adaptation Science Center (Postdoctoral Fellow)	2025 – Present

SELECT PRESENTATIONS

†Student mentee presenter • Total presentations: 41

12. **Baecher, JA. 2026.** *Ecology of rapid change: From behavior to movements to range dynamics.* **Emory University**, Atlanta, GA [[slides](#)] [[video](#)]
11. **Baecher, JA. 2026.** *Spatial structures for ecological models: A tutorial for point-referenced data.* **Utah State University** (hybrid) [[slides](#)] [[online resource](#)]
10. **Baecher, JA. 2025.** *Shifting Frontiers: Understanding species abundance range-shifts for informed conservation strategies.* **Pacific Flyway Partners** (hybrid) [[slides](#)]
9. **Baecher, JA. 2024.** *Influence of Global Change on Species' Behavior, Movement, and Ranges.* **SNRE Seminar Series**, Gainesville, FL [[flyer](#)]
8. **Baecher, JA, L Evans, BR Scheffers. 2024.** *Toward knowledge co-production in invasion modeling: Developing novel interactive expert-elicitation software.* **Invasive Species Research Institute Conference**, Gainesville, FL [[slides](#)]
7. **Baecher, JA, D Klings, L Evans, BR Scheffers. 2023.** *Informing Species Range-shifts with Global Connectivity.* **Species On The Move**, Ft Myers, FL [[slides](#)]
6. **Baecher, JA, L Evans, BR Scheffers. 2022.** *Predicting the spread of invasive Burmese pythons in Florida.* **SERISCC Network Webinar** (virtual) [[slides](#)] [[video](#)]
5. **Baecher, JA, SB Reichling. 2018.** *Current status of the Louisiana Pinesnake captive breeding program at the Memphis Zoo.* **USFWS Working Group Meeting**, Shreveport, LA
4. **Baecher, JA, SC Richter. 2018.** *Natural gradients influence the fine-scale distribution and abundance of salamanders in an old growth Appalachian forest.* **Joint Meeting of Ichthyology and Herpetology**, Rochester, NY
3. **Baecher, JA, P Vongrinc, JD Willson. 2016.** *Herpetofauna of degraded and restored remnant Tallgrass Prairie.* **Joint Meeting of Ichthyology and Herpetology**, New Orleans, LA
2. **Baecher, JA, DR Leasure, D Lynch, DD Magoulick. 2013.** *Impact of anthropogenic land use and hydrologic disturbance on crayfish assemblages.* **Joint Meeting of Aquatic Sciences.** Portland, OR
1. **Baecher, JA, J Ortega, SJ Beaupré. 2013.** *Use of stable isotope analysis for determining the diet breadth of the Western Cottonmouth (Agkistrodon Piscivorous).* **Society for Freshwater Science Annual Meeting**, Jacksonville, FL

WORKSHOPS

1. Fletcher, RJ, **JA Baecher**, M Lezzi, A Marx. *2023. Application of connectivity modeling to identify and predict movement and range redistribution. **Species on the Move**, Bonita Springs, FL [[slides](#)]*
2. **Baecher, JA**, L Evans, BR Scheffers. *2022. Expert elicitation of key invasive species habitat requirements, prevalence, and life history attributes. **Austin Cary Forest**, Gainesville, FL [[slides](#)]*
3. Fletcher, RJ, **JA Baecher**, M Lezzi, A Marx. *2022. Predicting the spread of invasive Burmese pythons in Florida. **Annual Meeting of the International Association of Landscape Ecologists - North America** (virtual) [[video](#)]*

TEACHING

Instructor of Record

University of Florida | School of Natural Resources and Environment
Environmental Science Lab

Fall 2022 & 2023

Teaching Assistant

Eastern Kentucky University | Department of Biology
Human Anatomy Lab

Spring & Fall 2015 – 2017

Guest Lectures (Selected)

Utah State University | Department of Wildland Resources
Landscape Ecology

Fall 2025

University of Florida | School of Natural Resources and Environment
Community Science for Ecologists

Spring 2025

University of Florida | School of Natural Resources and Environment
Ecology of Climate Change

Spring 2023

University of Florida | School of Natural Resources and Environment
Wildlife of Florida

Spring 2022

Rhodes College | Department of Biology
Vertebrate Biology

Spring 2023

Eastern Kentucky University | Department of Biology
Applied Learning in Biology

Spring 2016

Radford University | Department of Biology
Appalachian Herpetology

Spring 2016

University of Arkansas | Department of Biology Herpetology
Herpetology

Spring 2014

STUDENT MENTORSHIP

Graduate student (MS, PhD) • Φ published • Ψ conference • Ω committee member • $\rightarrow$ career prep

Eastern Kentucky University

2015 – 2017

Alex Gibson, Emily Jones (Φ , Ψ , $\rightarrow$ MS), Sarah Ashbrook (Ψ), Peyton Blair (Ψ), [Peter Grap] (Ψ , $\rightarrow$ MS, $\rightarrow$ PhD), Kelley Hoefer (Ψ), Myesha James (Ψ), [Jacob Peters] (Ψ , $\rightarrow$ MS, $\rightarrow$ PhD), [Julia Portmann] (Ψ , $\rightarrow$ MS, $\rightarrow$ PhD), [Jocelyn Stalker] (Φ , Ψ , $\rightarrow$ MS, $\rightarrow$ PhD)

Rhodes College

2018 - 2019

[Madeline Estes] (Ψ , Ω , $\rightarrow$ MS)

University of Florida

2019 – 2025

Brittany Arthur, Noah Bowen, Sierra Scauzillo (Ψ , Ω), [V Anand Akshay] (Φ , Ψ , PhD), [Amy Bauer] (Φ , Ψ , PhD), [Yasmin Tavares] (MS, Φ , Ψ , $\rightarrow$ PhD), [Yesenia Sanchez] (Φ , Ψ)

Utah State University

2025 – Present

Jack Rasmussen (PhD), [Mallory Sandoval-Lambert] (MS), [Adrien Bastidas] (PhD)

SELECT OUTREACH EDUCATION

Total Events: 26 • People Reached: ~1,430 (ages 5 to 60+)

Activity	Event	# Reached	Year
Guest teacher – ecosystems	Holt Middle School, AR	~30	2013
Watershed clean-up	Lake Fayetteville, AR	~30	2013
High school science fair judge	NWA STEM Fair, AR	~50	2013
Guest teacher – reptiles and amphibians	Model Elementary, AR	~15	2015
Environmental stewardship	EKU Career Fair, KY	~150	2016
Community water quality testing	Eastern Kentucky	~30	2016
Vertebrate conservation education	Memphis Zoo, TN	~300	2018

[[see all events](#)]

PUBLIC RESOURCES

Available at [[questlab.dev](#)]

Mixed Effects Models in R [

2025

Spatial Structures in Ecological Models | Tutorial [

2025

Intro to R Coding:

Part 1: the basics [

2022

Part 2: advanced techniques [

2022

Machine learning the “Tidy” way [

2021

Linear regression with gradient decent [

2021

Mapping ecological flow in R [

2020

Empirical Dynamic Models for forecasting [

2016

REFERENCES

Since PhD (2019 - Present)

Robert Guralnick, PhD 2019 – Present
Curator of Biodiversity Informatics | Florida Natural History Museum
352-273-1980 • [[email](#)] • [[webpage](#)]

Brett Scheffers, PhD 2019 – Present
Associate Professor | Dept of Wildlife Ecology, University of Florida
Director | Florida Climate Institute, University of Florida
352-846-0570 • [[email](#)] • [[webpage](#)]

Lindsay Campbell, PhD 2024 – Present
Assistant Professor | Florida Medical Entomology Lab, University of Florida
772-226-6666 • [[email](#)] • [[webpage](#)]

TJ Clark-Wolf, PhD 2025 – Present
Assistant Professor | Dept of Wildland Resources & Ecology Center, Utah State University
email preferred • [[email](#)] • [[webpage](#)]

Erica Stuber, PhD 2025 – Present
Assistant Unit Leader - USGS UT Cooperative Fish and Wildlife Research Unit
Assistant Professor - Dept. of Wildland Resources & Ecology Center
email preferred • [[email](#)] • [[webpage](#)]